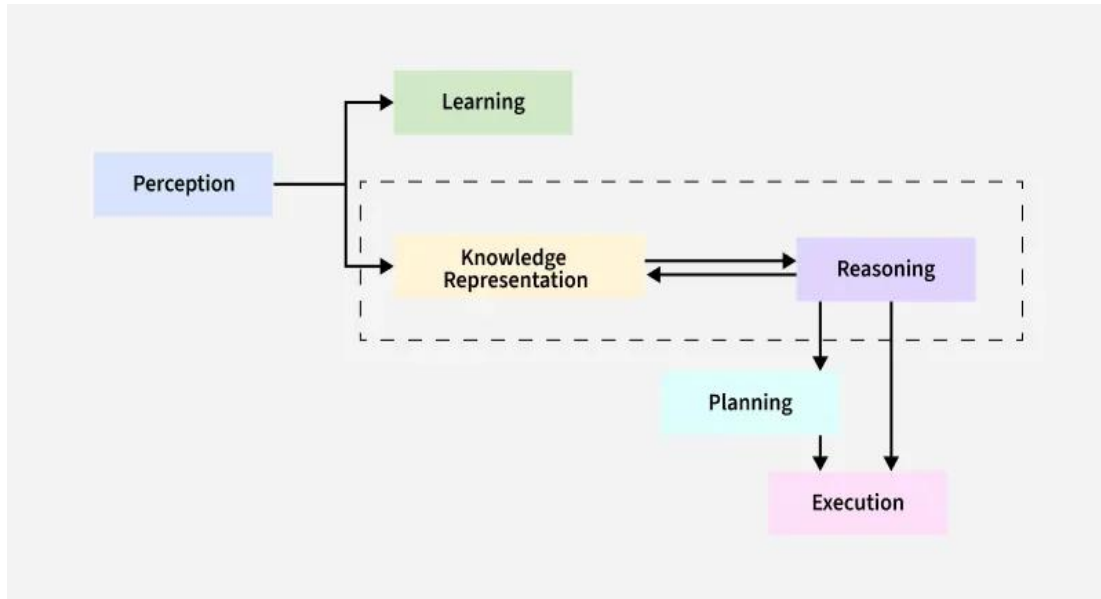


Introduction to Knowledge Representation in AI

Introduction



- **knowledge representation (KR)** in AI refers to encoding information about the world into formats that AI systems can utilize to solve complex tasks.
- This process enables machines to reason, learn, and make decisions by structuring data in a way that mirrors human understanding.
- Artificial intelligence systems operate on data. However, raw data alone does not lead to intelligence.
- AI must transform data into structured knowledge.
- KR achieves this by defining formats and methods for organizing information.
- With clear representations, AI systems solve problems, make decisions, and learn from new experiences.

The Role of Knowledge in AI

- AI needs knowledge to understand and respond accurately.
- Knowledge allows AI to function in real-world applications.

Example: AI chatbots need stored knowledge to answer questions correctly.





Why is Knowledge Representation Important?

- It helps AI understand and process information.
- It allows AI to make better decisions.
- It makes human-computer interaction easier.

Example: AI assistants like Siri and Google Assistant use KR to answer questions.

Characteristics of Good Knowledge Representation

- **Easy to Understand:** The stored knowledge should be clear and structured.
- **Efficient:** AI should be able to find and use knowledge quickly.
- **Expandable:** New information should be easy to add.
- **Accurate:** The knowledge should be correct and logical.

Types of Knowledge

- **Simple Relational Knowledge:** This is like organizing facts neatly in columns, often used in databases. It's straightforward but not great for drawing conclusions. For instance, in a database, you can use this to list relationships between people and their addresses.
- **Inheritable Knowledge:** Here, data is stored in a hierarchy, like a family tree. For example, you can use this to show how animals relate to different species or how products belong to various categories. It helps us understand relationships between things, and it's better than the simple relational method.
- **Inferential Knowledge:** This is the precise way of using formal logic to guarantee accurate facts and decisions. For instance, you can use this to deduce that if "All men are mortal" and "Socrates is a man," then "Socrates is mortal."
- **Procedural Knowledge:** AI uses small programs or rules (like recipes) to perform tasks. For example, it can follow rules to play chess or diagnose diseases. Despite its limitations, it is useful for specialized tasks.

Types of Knowledge Representation

- 1) **Logical Representation:** Uses rules and statements (like math formulas).
- 2) **Structural Representation:** Uses diagrams to show relationships. (Semantic nets, frames)
- 3) **Probabilistic Representations:** (Bayesian networks, Markov decision process)

Logic-Based Representation

Logic-based methods use formal rules to model knowledge. These systems prioritize precision and are ideal for deterministic environments.

Propositional Logic:

- Represents knowledge as declarative statements (propositions) linked by logical operators like AND, OR, and NOT.
- For example, "If it rains (A) AND the ground is wet (B), THEN the road is slippery (C)." While simple, it struggles with complex relationships. Often follow the format "IF condition THEN conclusion."
- For instance, in a knowledge-based system, you might have:
- IF an object is red AND round, THEN the object might be an apple.

First-Order Logic (FOL):

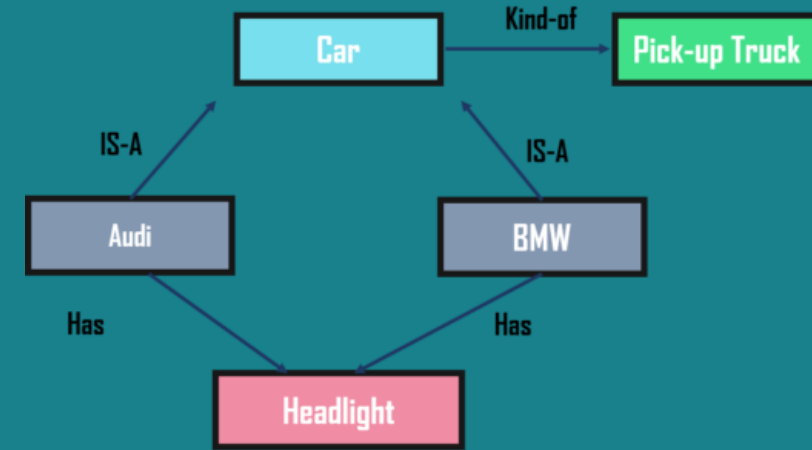
- Extends propositional logic by introducing variables, quantifiers, and predicates. FOL can express statements like, "All humans ($\forall x$) are mortal ($\text{Mortal}(x)$)." It supports nuanced reasoning but demands significant computational resources.
- Legal AI tools apply logic-based rules to analyze contracts for compliance.

Structured Representations

These methods organize knowledge hierarchically or through networks, mimicking how humans categorize information.

Semantic Networks:

- Represent knowledge as nodes (concepts) and edges (relationships).
- For example, "Audi" links to "Car" via an "Is-A" connection. They simplify inheritance reasoning but lack formal semantics.



Frames:

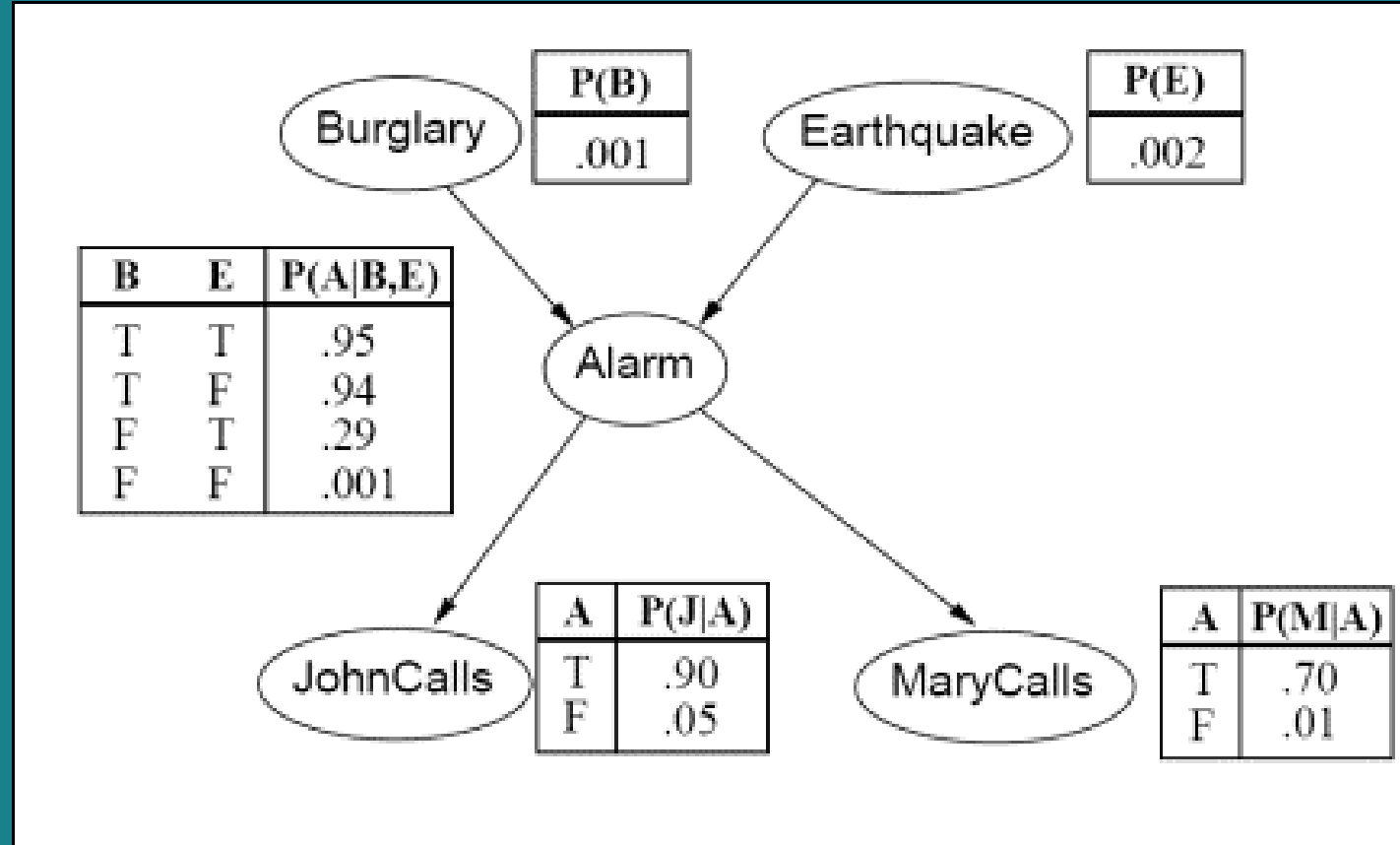
- Frames are data structures for representing stereotyped situations, similar to objects in object-oriented programming. A "Vehicle" frame may include slots like wheels, engine type, and fuel. Frames excel in default reasoning but struggle with exceptions.

Ontologies:

- Formal representation of a set of concepts within a domain and the relationships between them. It is used to create a shared understanding of a domain.
- Example: In a medical ontology, "Disease" might be connected to "Symptom", "Treatment", etc.

Probabilistic Models

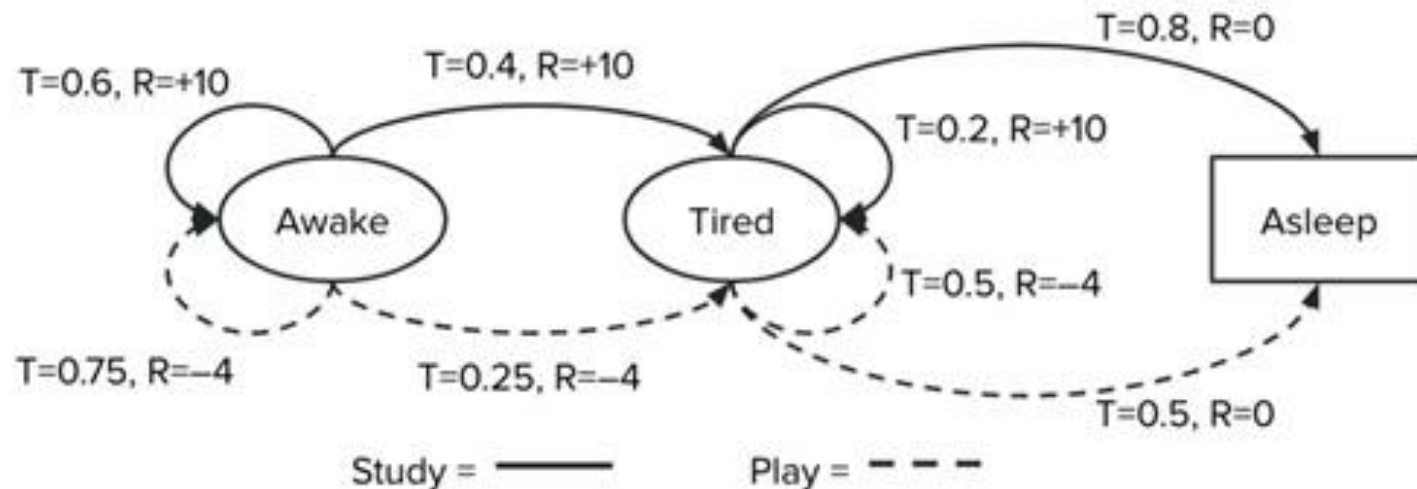
- These systems handle uncertainty by assigning probabilities to outcomes.
- **Bayesian Networks:**
- Use directed graphs to model causal relationships. Each node represents a variable, and edges denote conditional dependencies.
- For instance, a Bayesian network can predict the likelihood of calling John or Mary based on alarm event.



Probabilistic Models

- **Markov Decision Processes (MDPs):**
- Model sequential decision-making in dynamic environments. MDPs help robotics systems navigate obstacles by evaluating potential actions and rewards.
- Weather prediction systems combine historical data and sensor inputs using probabilistic models to forecast storms.

The Markov Decision Process below models three states – Awake, Tired, and Asleep – and two actions – Study (the solid arrows) and Play (the dashed arrows). The transition value (i.e. probability) and reward for each (s, a, s') triple is provided on the arc of each arrow. The Asleep state is a terminal state with no actions. (Alternatively, you could think of the Asleep state as having a single non-action arc looping back to Asleep with probability 1.0 and reward 0.)



Challenges in Knowledge Representation

- **Complexity:** Representing the full richness of human knowledge is a daunting task, and representing ambiguity or incomplete knowledge is difficult.
- **Scalability:** As knowledge bases grow, reasoning can become computationally expensive.
- **Uncertainty:** Real-world knowledge is often uncertain, vague, or contradictory.
- **Common Sense Knowledge:** Capturing basic knowledge that humans take for granted is still a significant challenge for AI systems.

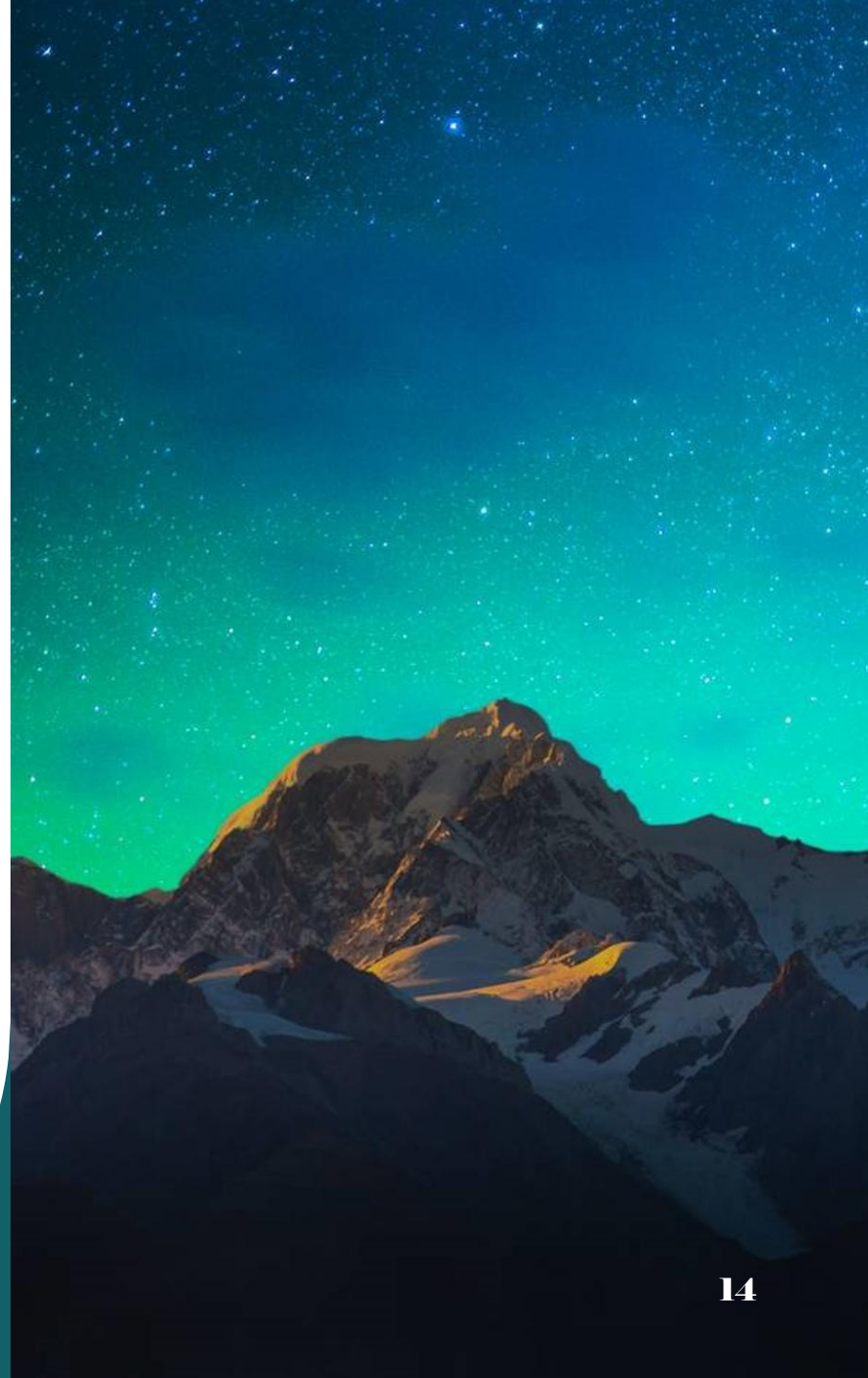
How AI Learns from Knowledge Representation

- AI learns from knowledge representation by structuring information in a way that allows it to interpret data, draw inferences, and reason through complex problems
- It is mimicking human thought processes to make decisions and solve tasks based on the knowledge it has been given, similar to how a human would use their understanding of the world to navigate situations.
- This enables AI to go beyond simple data analysis and apply deeper understanding to new scenarios.
- AI also uses machine learning models in order to make predictions using data.

Example: AI using KR to predict weather conditions.

Where is Knowledge Representation Used?

- **AI Assistants** (Siri, Alexa, Google Assistant)
- **Medical Diagnosis** (AI helping doctors)
- **Chatbots** (AI answering questions online)
- **Self-driving Cars** (AI making driving decisions)



The Future of Knowledge Representation

- **AI will continue to improve how knowledge is stored and used.**
- **KR will evolve to handle more complex and dynamic information.**
- **Better KR systems will lead to more advanced AI applications.**

Summary

- Knowledge Representation helps AI store and use information efficiently.
- There are different ways to represent knowledge, like logic, networks, frames, rules, and hierarchies.
- It is widely used in AI applications that we see every day.

Thank you